

Value Leakage in Frequent Flyers Program Analysis

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Data and methodology

1 Data Source

An airline loyalty program data from a fictitious canadian-based airline that held a sign-up promotion in 2018.

Source: Maven Analytics.
Total rows: 400K+

2 Data Processing

The data is aggregated and filtered by travelers who signed up in 2018 only to reduce tenure bias with Python.

3 Statistics Test

A significance test is done to find out whether the number is statistically significant.

4 The Flyer Points Leakage

A leakage happens when the median of dollar points redeemed exceeded CLV.

Resulting in the promotion's failure to bring high-valued travelers, and make revenue for the airline.

While the 2018 promotion drove rapid acquisition, some challenges remained unresolved.

1 Massive User Acquisition

The campaign aggressively expanded the Frequent Flyer base. But their quality remained unrevealed.

2 Churn Definition

Standard churn report may miss ‘Silent Churn’, travelers who just stop flying without explicitly unsubscribing to the program.

Real churn = Silent Churn + Explicit Churn.

3 The Discount Travelers

Did the airline subsidize low-value travelers who artificially inflate engagement metrics like signed up travelers?

4 Financial Leakage

Are we paying massive points redemption to users who don’t generate positive net lifetime value?



To solve the challenges, we tested the 2018 cohort across three phases to uncover the campaign's true ROI.

#1 The Hidden Loss

How many travelers abandoned the airline when accounting for invisible drop-offs? (silent churn)

#2 Promo Quality

Did we acquire sustainable high-value flyers or transient discount hunters?

#3 Points Leakage

Did the dollar cost of points redeemed cannibalize the lifetime value of the customer?

The 2018 Promotion successfully acquired highly engaged, profitable travelers with zero financial leakage

17% Churn Rate

Promo flyers exhibited a lower ‘Real Churn’ rate than Standard, but is **statistically insignificant**.

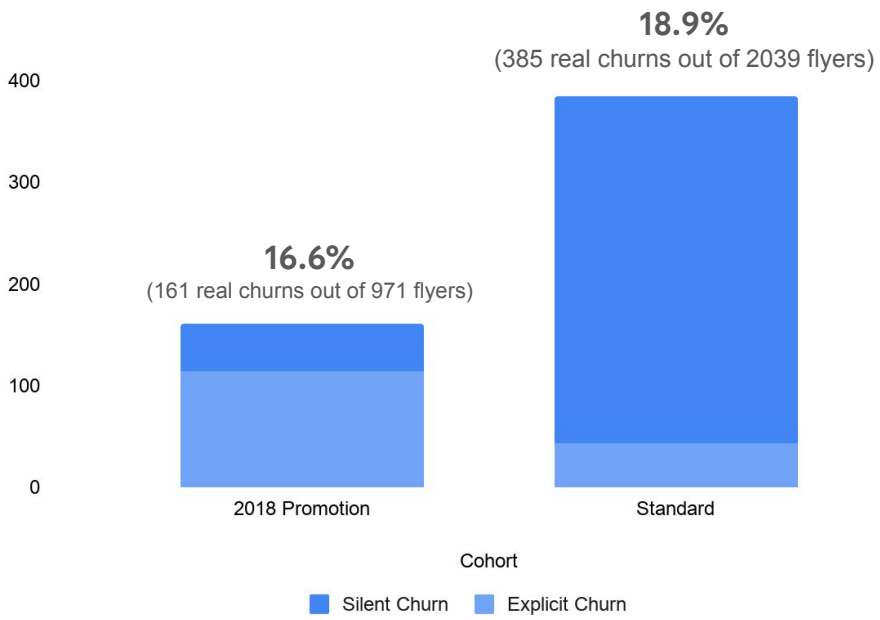
7x more flights

Promo travelers flew around 7x more often with a CLV almost equal to standard travelers.

0.0% Leakage

0.0% of promo travelers caused financial leakage. Their redemption costs were absorbed by net lifetime value.

#1 Users acquired via the 2018 Promotion exhibited 2.3% lower real churn, but statistically insignificant.



Key takeaway: The promotional campaign captured a positive signal where the promotion group retained 2% higher travelers, but inconclusive.

2.3% difference, how significant?

A Two-proportion Z-Test yields a p value of **0.1257 (>α 0.05).**

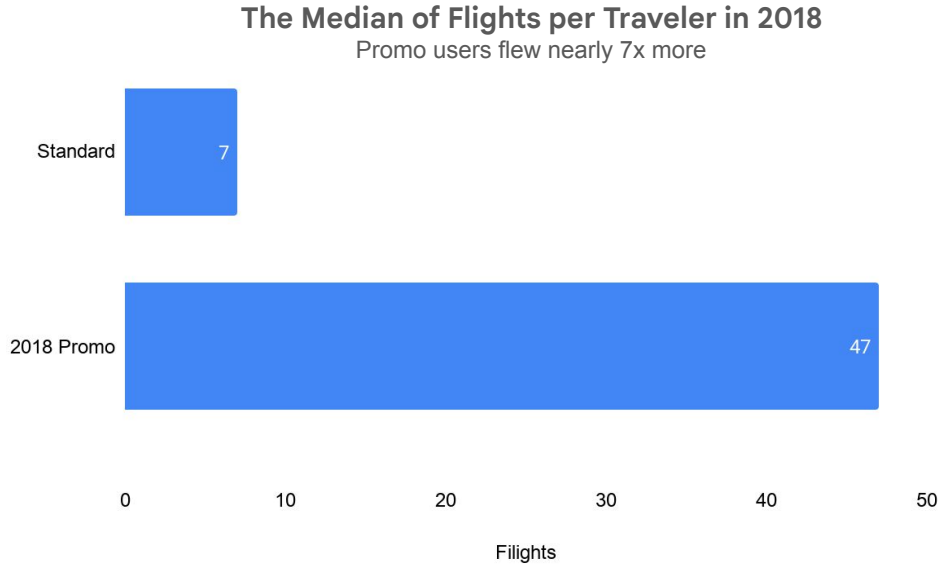
Meaning the difference is not significant.

#2 Promo travelers tend to fly 7x higher while maintaining an on par CLV with standard travelers.

Travelers CLV

The median CLV for promo travelers were **C$5,926** compared to **C$5,575** standard travelers

Significance: Hypothesis testing confirms the CLV difference between groups is insignificant (p = 0.58)



The promotion did not weaken travelers quality, despite showing neutral significance



#3 Financial leakage happens when the points redeemed exceed traveler's lifetime value.

Customer Lifetime Value (CLV)
(The total revenue of travelers in 2018)

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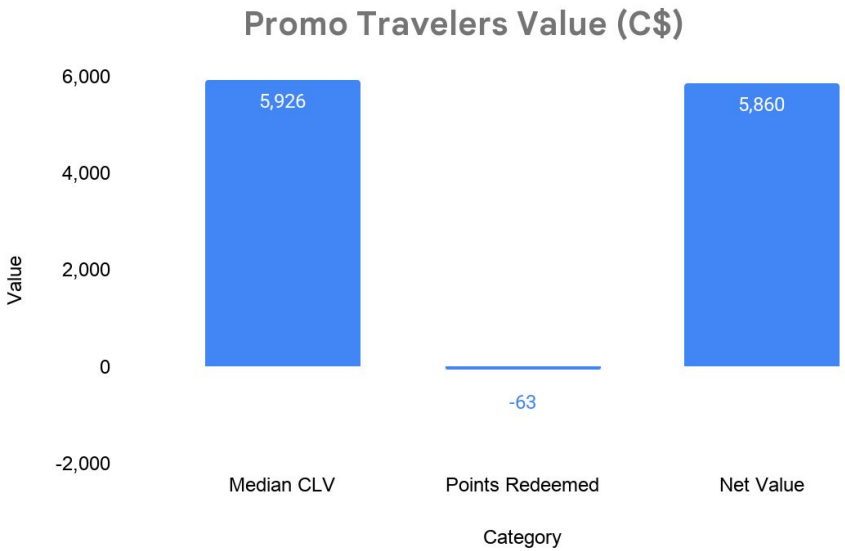
Dollar Cost of Points Redeemed
(The C$ cost paid by the airline)

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Net Value
(The profitability of user)

A leakage happens when Net Value <0.

#3 Although promo travelers redeemed more points, their inherent CLV ensured zero financial leakage for the airline.



Key takeaway: Out of 971 Frequent Flyers who signed up during the 2018 promotion, their points redemption did not cause leakage.

The median of points redeemed were absorbed by the travelers' CLV (5,926 - 63 = 5,860)

Across three phases, the 2018 promotion signaled a positive result for profitable traveler acquisition strategy.

	Standard	2018 Promotion
Real Churn Rate	18.9%*	16.6%*
Median Flights/Yr	7	47
Median CLV	C$5,575*	C$5,926*
Leakage Rate	0.0%	0.0%

**indistinguishable, statistical parity*

Key takeaway: Experiment further with the mechanics of the 2018 campaign to drive sustainable future growth

Update Reporting

Standardize the ‘Silent Churn’ (12+ month of inactivity) to ensure accurate calculation

Experiment The Success

The 2018 promo did not trigger massive churn rate, and signaled a healthy net value. The airline may **experiment** with similar campaign mechanics.

Reframe Redemption

Shift focus from ‘reducing redemption cost’ to ‘maximizing engagement’. The analysis signals high redemption correlates with profitable, engaged travelers.



Appendices

Google Colab Link
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